FIGURE 01 - CHANGEMAKERS TEAM

AUTHOR'S ARCHIVE

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**Mechanical Engineer**
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**Designer**

# CHANGEMAKERS

## São Paulo, Brazil Soccer Infrared – ROBOCUP JR.

## 1.0 ABSTRACT

We are the **SESI ChangeMakers team** from **SESI São Paulo, Brazil**, and we are currently in **our second season** competing in **RoboCup Junior Soccer Infrared.** We have been continuously working on the **design of our current robots** for **approximately one year:** we began the planning process in **November 2025**, focusing on improvements in **hardware, software, mechanics, and system integration**, with the goal of increasing **reliability, efficiency, and performance during matches**, as well as developing our **own solutions** that are **replicable** and can be **continuously improved.**
Regarding **the mechanical design**, the chassis was **entirely designed by the team** and manufactured from **ABS using 3D printing**, aiming to achieve a balance between structural **strength**, **weight reduction**, and **dynamic performance**. The design prioritizes **modularity**: it is composed of four modules, facilitating **maintenance**, **component replacement**, and the **implementation of future improvements.**
The control system, **developed in C++**, was **structured in a modular and flexible manner**, facilitating **development**, **maintenance**, and **integration** among the **different onboard modules**. The **electronic architecture** was also improved: we designed our own **Printed Circuit Boards (PCBs)**, which are one of the **main distinguishing features of the project**. The ***upper, main, kicker, and lower board*** were developed providing more **efficient** component organization, **greater reliability** in **electrical connections**, and better integration among the **robot's systems.** Information regarding team contact details, files, electrical schematics, and CAD models is available through the links in the **footer of the banner.**

## 2.0 HARDWARE

### *ENGINEERING AND MANUFACTURING DESIGN (MECHANICAL AND ELECTRONICS)*

### *2.1 MECHANICAL*

**In the previous season, the robot's chassis was constructed from PMMA (acrylic). For the current season, we opted to use ABS, aiming to reduce mass while maintaining greater mechanical strength against the loads experienced during matches. The material selection was evaluated through CAE stress simulations in Autodesk Inventor, comparing both materials under the same conditions and using the same geometry.**
**For the analysis, a 500 N force was applied perpendicular to the chassis. This value was derived from a simulated frontal collision between two 1.5 kg robots, moving at an approximate speed of 1.57 m/s. This speed was estimated from the robot's actual drivetrain configuration, using four 500 RPM motors and 60 mm diameter wheels, representing a realistic high-speed collision scenario that can occur during a match. The same load position, direction, and magnitude were applied to both the PMMA and ABS models, allowing a direct comparison of stress, strain, and displacement. This analysis helped validate the use of ABS based on the actual robot geometry, rather than solely on its theoretical material properties, while also considering the need for low mass and mechanical resistance.**
**The mechanical development included four chassis prototypes/versions, with progressive changes in material, structure, and modularity. The final robot has a mass of 1500 g and dimensions of 215 mm × 215 mm.**

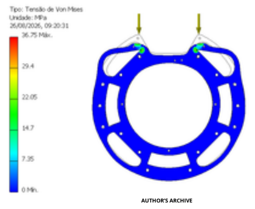

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**FIGURE 02 – DEFORMATION OF THE PMMA STRUCTURE**

To evaluate **the structural** resistance of the **PMMA (Figure 02)** used in the **previous version** of the robot, a simulation was performed in **Autodesk Inventor**, applying **a 500 N force** to the front part of the **lower plate**. The simulation indicated approximately **45% deformation** in the most **critical region of the structure.** This result highlighted the structural **limitations of PMMA** when subjected to a concentrated load and contributed to the **evaluation of its suitability for the robot chassis.**

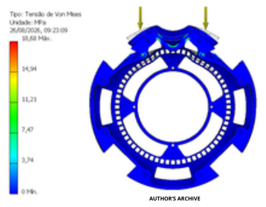

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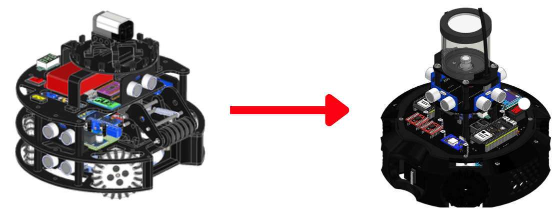
**FIGURE 03 – DEFORMATION OF THE ABS STRUCTURE**

The **ABS chassis (Figure 03)**, used in the **current version**, was evaluated under **the same 500 N force** in **Autodesk Inventor**. The simulation indicated **approximately 15%** deformation, demonstrating a **significant reduction** compared to the **PMMA structure.** This represents an approximately ***67%*** reduction in deformation, while **maintaining the structural integrity of the chassis.** The results supported the **selection of ABS as a more suitable structural material,** providing greater resistance to deformation and **better performance under the mechanical demands** expected during matches.

### *• INNOVATION*

One of the **main evolutions** of the project was the **replacement of acrylic sheets** with **PCBs** (Printed Circuit Boards) as **structural elements**, integrated into the chassis architecture. This improved the **organization** of the **modules** and promoted a **reduction in wiring**, resulting in a more **compact**, **robust** system that is **less susceptible to failures**. In addition, there was a **reduction in mass**, **better distribution** of mechanical loads, and **improvements** in **assembly**, **maintainability**, and **modularity**.

FIGURE 04 - COMPARISON BETWEEN ROBOTS FROM DIFFERENT SEASONS



*2025 SEASON ROBOT (FABULOSO)* *2026 SEASON ROBOT (NEXUS)*

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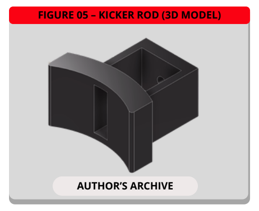
### *• HIGHLIGHTS*

FIGURE 05 – KICKER ROD (3D MODEL)



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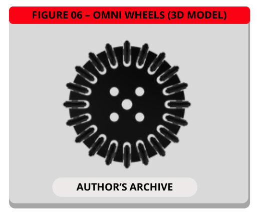
As a **mechanical highlight**, the **kicker rod (Figure 05)** was developed with a **specific** curvature, which facilitates the **capture** and **directioning** of the **ball** during the robot's movement. The switch coupled to the rod allows the mechanism to be **activated in an integrated manner**, while its **curved shape** contributes to **better handling** and **ball control**, reducing the possibility of **losing the ball** during movements and changes in direction.

FIGURE 06 – OMNI WHEELS (3D MODEL)



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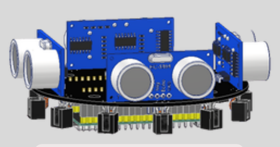
The robot uses an **omnidirectional wheel** system **(Figure 06)**, which enables **independent movement** along the **X** and **Y axes**, as well as rotation around **its own axis**. This configuration allows **lateral**, **longitudinal**, and **diagonal** movements **without** the need to **reorient** the chassis. The separation of motion-axis control provides greater **positioning accuracy**, **better dynamic response**, and **greater control** during rapid maneuvers on the field, directly contributing to the robot's **efficiency** in **offensive** and **defensive** situations.

### *2.2 ELECTRONICS*

#### ① UPPER BOARD

For this season, an **upper board** was developed to **integrate** the system's **main sensors** (Figure 07). **Twelve TSSP4038 sensors** were mounted on the **underside** of the plate, with the function of **detecting the infrared signals** emitted by the ball. On the same plate, on the **upper side**, **four HC-SR04 ultrasonic sensors** were integrated and arranged in a **cross** configuration, with the purpose of locating the robot on the field by **identifying walls** and the **goal** through **distance measurements**, along with an **ESP32-S3** for **processing**. In addition, an **OpenMV Cam RT1062** camera was integrated, operating together with a **convex mirror**, enabling **omnidirectional vision** of the environment. This system allows the **identification of the field** and the **ball** through **image processing**, contributing to the **improvement** of **game strategies**.

- **FIGURE 07 – UPPER BOARD (3D MODEL)**



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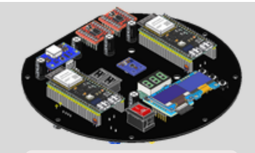
#### ② MAIN BOARD

The **main board** was designed to serve as the **control** and **integration** core of the robot's **electronic system** (Figure 08), concentrating the main **processing**, **communication**, and **actuation modules**, which ensures **efficient interconnection** between the different subsystems. The circuit features **dedicated connectors** for the **kicker** board, **upper** module, and **lower** module, enabling **organized communication** while also facilitating **maintenance** and **future system expansions**. For actuator control, **two TB6612FNG motor drivers** were integrated, responsible for the **precise** and **efficient** control of the drive motors. The robot's orientation is managed through a **BNO055 compass sensor**, which provides **directional information** and contributes to **navigation** and **stabilization** on the field. Communication with the official competition systems is ensured through the integration of the **RCJ** (RoboCup Junior) **module**. In addition, the system incorporates an **OLED display**, used to demonstrate **diagnostic information**, **operational states**, and robot **parameters**, assisting with **testing** and **adjustments**. Processing is carried out by **two ESP32-S3** microcontrollers (the **actuator** and the **processor**), which operate in a **distributed** manner, **optimizing performance** and **communication** between the electronic modules. Power is supplied through a battery connector, with the voltage regulated by a **Mini 560 Step-Down converter**, ensuring a **stable**, **efficient**, and **suitable** power supply for **all** system components.

- **FIGURE 08 – MAIN BOARD (3D MODEL)**



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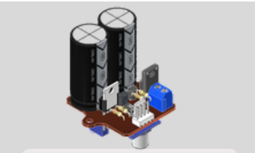
#### ③ KICKER BOARD

The **kicker board** was designed for the **controlled storage** and **release of energy** (Figure 09), enabling the **efficient** execution of the **ball kick**. The system is powered by **12 V** from the main battery and uses an **XL6009 boost converter** to raise the voltage to an appropriate level for **charging** the power circuit. The energy is stored in a bank of **two 2200 µF capacitors**, capable of delivering **high currents** over a **short period of time**. The trigger command operates at 3.3 V and is electrically isolated by a **PC817 optocoupler**, **protecting** the control electronics against **noise** and **voltage spikes**. The control signal is **amplified** by a **transistor** stage, responsible for the **rapid** activation of the **IRLZ44N MOSFET**, which acts as an electronic switch for the **instantaneous** discharge of the capacitors into the **solenoid**. The high current generates the **magnetic field** required to **propel the plunger** and **perform the kick**. After firing, the system **automatically recharges**, ensuring **readiness** for subsequent actions.

- **FIGURE 09 – KICKER BOARD (3D MODEL)**



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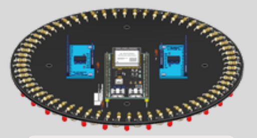
#### ④ LOWER BOARD

The **lower board** was developed for the **field line detection system** (Figure 10). The assembly uses **32 LDR** sensors combined with **32 red LEDs**, responsible for the controlled **illumination** of the surface and the **stability** of the measurements. Each sensor and each light emitter has a **dedicated resistor**, totaling **64 resistors**, ensuring the **correct electrical operation** of the circuit. Signal processing is performed by an **ESP32-S3** microcontroller, responsible for the **acquisition** and **processing** of information. The **readings** are obtained through **two CD74HC4067 multiplexers**, allowing for **expanded inputs** and **system organization**. The **identification** of the markings is based on the difference in **reflectance** between the **white of the lines** and the **green of the field**, with the use of red LEDs to increase **contrast**. The processed information is then transmitted to the second ESP32-S3, located on the main board, where it is **integrated** into the overall **control logic** of the robot.

- **FIGURE 10 – LOWER BOARD (3D MODEL)**



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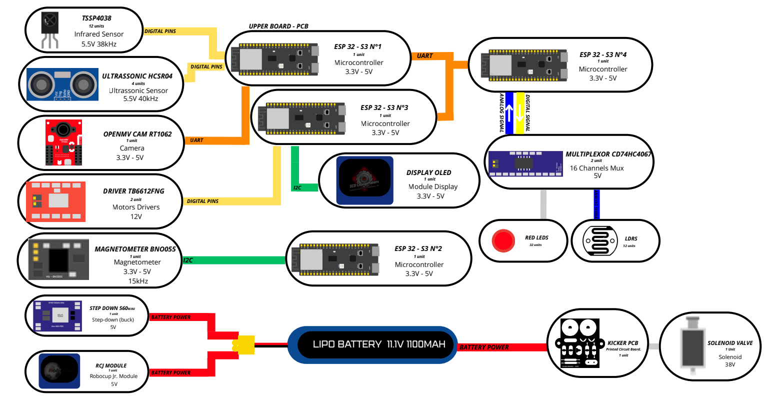
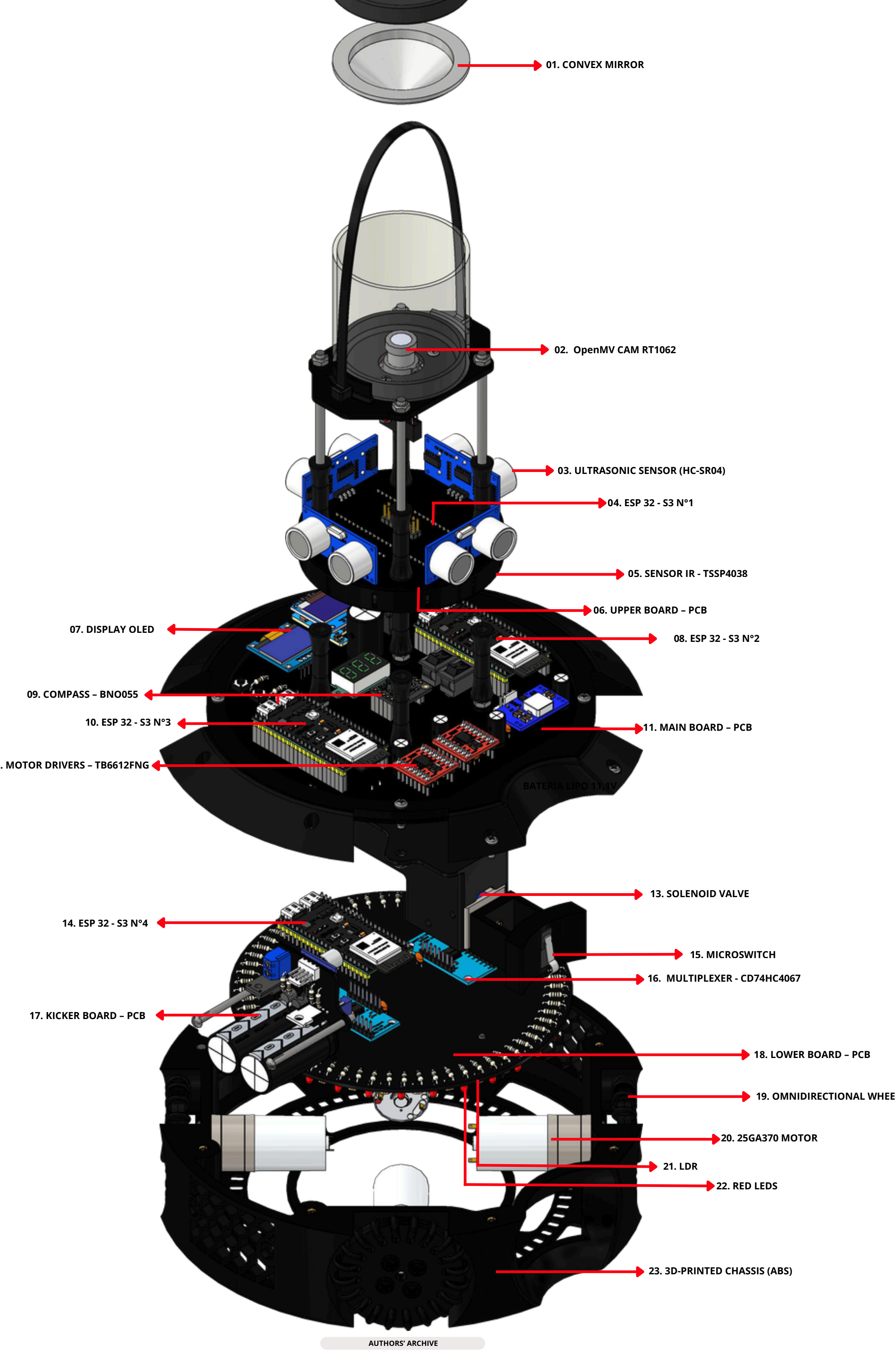


### *• ELECTRICAL ARCHITECTURE*

The **electrical schematic** presents, **in detail**, the organization and interconnection of the **main electronic components** present in the robot. The **diagram** represents the **connections** between the different **ESP32-S3s**, **sensors**, **motor drivers**, **power supply systems**, and **other circuits**, allowing an understanding of how the modules are **integrated** and how **communication** between them occurs. In addition to representing the electrical connections, the schematic was used as a **reference** during the development and assembly of the boards, assisting in the identification of **inputs**, **outputs**, **power lines**, and **communication signals**. The organization of the components into different modules also facilitates **maintenance**, the **identification of potential failures**, and the implementation of **modifications** to the system. Thus, the schematic provides a more **organized** and **reliable** architecture, facilitating the **integration** and **operation** of the robot's different subsystems.

🟥 BATTERY POWER 🟨 DIGITAL PINS 🟨 DIGITAL SIGNAL 🟧 UART 🟩 I2C 🟦 ANALOG SIGNAL

## FIGURE 11 – 3D EXPLODED VIEW OF THE ROBOT



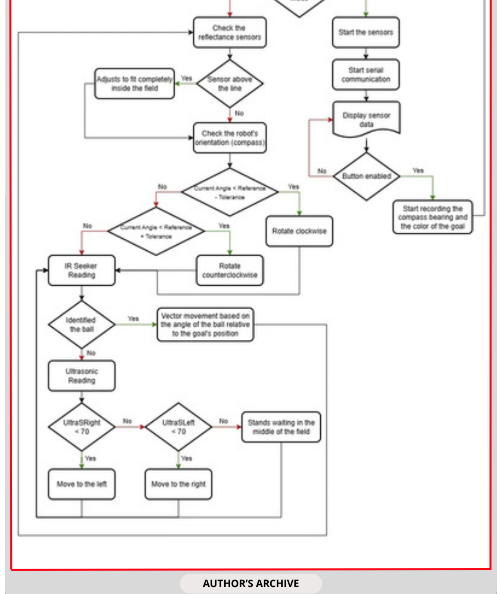
AUTHORS' ARCHIVE

**TOTAL ROBOT COST: $935** *(American Dollar)*
The stated amount corresponds to the manufacturing cost of ***one complete robot.***

## *3.0 SOFTWARE*

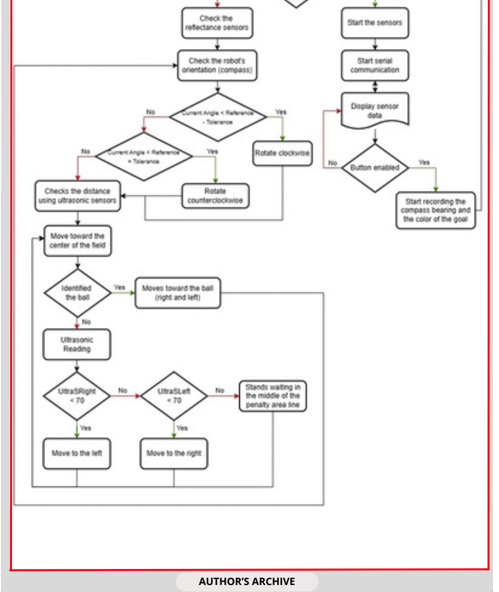
### *3.1 ROBOT STRATEGY ORGANIZED INTO FLOWCHARTS*

FIGURE 12 – ATTACKING ROBOT STRATEGY FLOWCHART


Start
Calibration Mode
No
Yes
Check the reflectance sensors
Start the sensors
Sensor above the line
Yes
Adjusts to fit completely inside the field
No
Start serial communication
Display sensor data
Check the robot's orientation (compass)
Button enabled
No
Yes
Start recording the compass bearing and the color of the goal
Current Angle < Reference - Tolerance
Current Angle < Reference + Tolerance
Rotate clockwise
Rotate counterclockwise
IR Seeker Reading
Identified the ball
Yes
Vector movement based on the angle of the ball relative to the goal's position
No
Ultrasonic Reading
UltraSRight < 70
UltraSLeft < 70
Stands waiting in the middle of the field
Move to the left
Move to the right

AUTHOR'S ARCHIVE

FIGURE 13 – DEFENDING ROBOT STRATEGY FLOWCHART


Start
Calibration Mode
No
Yes
Check the reflectance sensors
Start the sensors
Check the robot's orientation (compass)
Start serial communication
Display sensor data
Current Angle < Reference - Tolerance
Rotate clockwise
Current Angle < Reference + Tolerance
Button enabled
Start recording the compass bearing and the color of the goal
Checks the distance using ultrasonic sensors
Rotate counterclockwise
Move toward the center of the field
Identified the ball
Moves toward the ball (right and left)
Ultrasonic Reading
UltraSRight < 70
UltraSLeft < 70
Stands waiting in the middle of the penalty area line
Move to the left
Move to the right

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## 4.0 OUTREACH AND RCJ COMMUNITY ENGAGEMENT

*FIGURE 14 - OUTREACH AND COMMUNITY ENGAGEMENT*

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Throughout the season, **we worked to share our knowledge** and contribute to the **RoboCup community.** We developed a **beginner's guide available** on our GitHub, **bringing together information, tips, and practical guidance** to support students and teams taking their first steps in competitive robotics. We also supported the **Blue Birds and Red Rabbit (FLL) teams,** sharing our experience and helping them **overcome technical challenges.** Through this interaction, **both teams developed an interest in RoboCup Junior.** In addition, we participated in **SENAI's Technology Week**, where we presented our **robot and its technologies** to multiple groups of students from different areas of the institution. These presentations **allowed us to demonstrate our robot,** explain our solutions, and discuss our experience in **RoboCup Junior** and competitive robotics. Through these initiatives, we expanded access to **practical knowledge**, encouraged **interest in RCJ**, and **connected with students and teams** beyond our own competition environment.

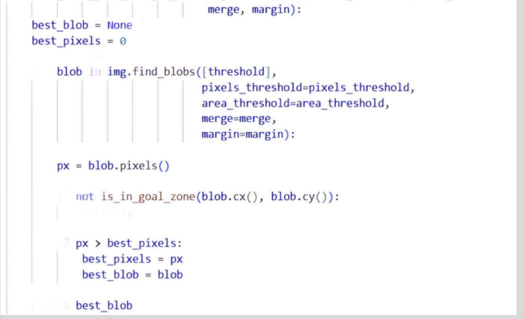
## 5.0 TOOLS USED

| | |
|---|---|
| **ALTIUM** | Robot Electronics Design |
| **AUTODESK INVENTOR** | Robot 3D Modeling |
| **EASYEDA** | Robot Electronics Design |
| **GITHUB** | Data Sharing and Storage |
| **OPENMV** | Camera Programming Software |
| **ULTIMAKER CURA** | 3D Modeling & Slicing Software |
| **VISUAL STUDIO CODE** | Robot Programming Software |

## 6.0 COMPUTER VISION OF THE FIELD

The **find_best_goal_blob_in_zone** function **(Figure 15)** implements a **generic segmentation** and object selection routine, applicable to **all elements identified** by the **vision system**, such as the **ball** and **goal**. Detection is performed using **color thresholds**, **area parameters**, and **pixel count**, in addition to **spatial filtering** by region of interest. As a **modular** and **reusable** function, its application to different elements only requires the **adjustment of parameters**, such as color **thresholds** and the **analyzed region**. This avoids code duplication and contributes to a more **organized**, **efficient**, and **scalable** system.

FIGURE 15 – CODE SNIPPET FOR COLOR THRESHOLD DETECTION



```
def find_best_goal_blob_in_zone(img, threshold, pixels_threshold, area_threshold,
                                merge, margin):
    best_blob = None
    best_pixels = 0

    for blob in img.find_blobs([threshold],
                               pixels_threshold=pixels_threshold,
                               area_threshold=area_threshold,
                               merge=merge,
                               margin=margin):

        px = blob.pixels()

        if not is_in_goal_zone(blob.cx(), blob.cy()):
            continue

        if px > best_pixels:
            best_pixels = px
            best_blob = blob

    return best_blob
```

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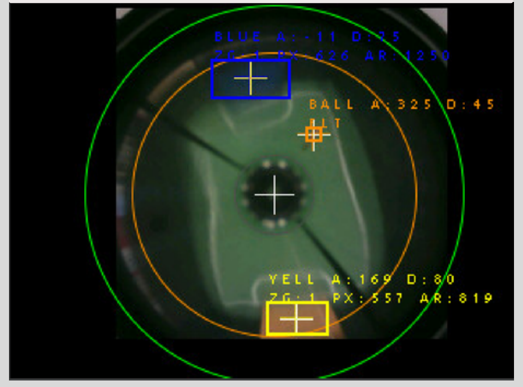
FIGURE 16 – ROBOT IMAGE WITH ALL CAMERA PARAMETERS



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*CAPTION:*

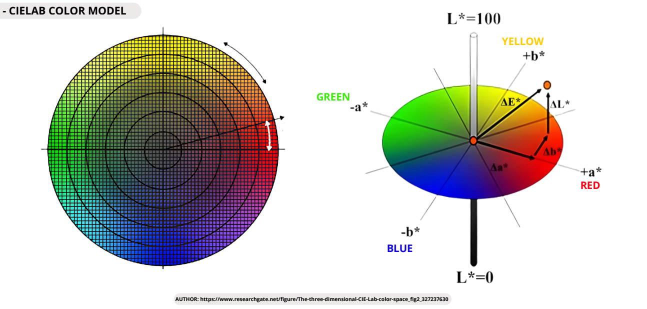
- *BLUE GOAL*
- *BALL*
- *YELLOW GOAL*

The **camera** captures the environment in **real time**, identifying the **ball** and **goals** for the robot's **decision-making**. Before each match, we ***calibrate*** the **sensor thresholds** during the **pre-match** period **to adapt the robot** to the current field and lighting conditions.

### *• 6.1 APPLICATION OF THE CIELAB MODEL IN THE VISION SYSTEM (L*A*B)*

The vision system uses the **CIELAB color model** (Figure 17) to perform **robust segmentation** of field elements. Unlike spaces such as **RGB**, CIELAB separates **luminance** information (L*) from the **chromatic components** (a* and b*), allowing **greater** color detection **stability** under different **lighting** conditions. To ensure consistency, a **white balance** process is applied, correcting deviations caused by variations in ambient light. In addition, adjustments to the L* component are used to **compensate** for changes in **brightness**, maintaining **reliability** in **color identification**. As a result, detection, especially of elements such as the yellow goal, becomes **more accurate** and **less sensitive** to **noise** and **external** variations.

FIGURE 17 - CIELAB COLOR MODEL



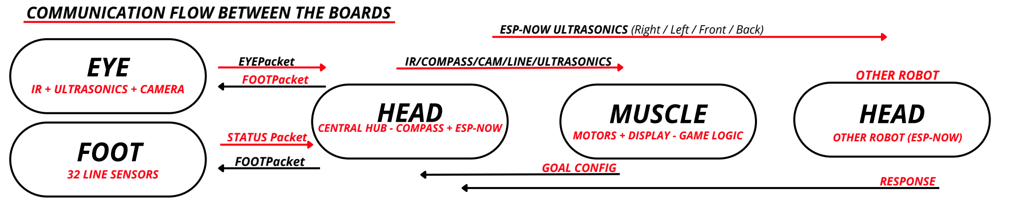
AUTHOR: https://www.researchgate.net/figure/The-three-dimensional-CIE-Lab-color-space_fig2_327237630

## 7.0 COMMUNICATION ARCHITECTURE AND DATA PACKAGES – INNOVATION

### *• 7.1 COMMUNICATION ARCHITECTURE*

The robots' communication system uses a **distributed architecture** across multiple **specialized boards**, where each module performs a **specific function** within the robot. This division provides **greater modularity**, **reduces coupling** between **sensors** and **actuators**, and improves **system stability** in real game situations. In addition, the separation of responsibilities facilitates **maintainability**, **debugging**, and the **independent evolution** of the firmware, increasing the **reliability** and **scalability** of the embedded architecture.
The exchange of information between the modules occurs through **structured packages**, ensuring **fast**, **organized**, and **synchronized** communication between the robot's different subsystems.

*COMMUNICATION FLOW BETWEEN THE BOARDS*


ESP-NOW ULTRASONICS (Right / Left / Front / Back)
EYE
IR + ULTRASONICS + CAMERA
EYEPacket
FOOTPacket
IR/COMPASS/CAM/LINE/ULTRASONICS
HEAD
CENTRAL HUB - COMPASS + ESP-NOW
MUSCLE
MOTORS + DISPLAY - GAME LOGIC
OTHER ROBOT
HEAD
OTHER ROBOT (ESP-NOW)
FOOT
32 LINE SENSORS
STATUS Packet
FOOTPacket
GOAL CONFIG
RESPONSE

*ESP-NOW TIMING*

PERIODIC TRANSMISSION: 2,000 ms | LINK TIMEOUT: 6,000 ms | STATUS -> MUSCLE: 500 ms | INIT RETRY: 2,000 ms | STATUS -> EYE/FOOT: 700 ms

### *7.2 BINARY FRAME STRUCTURE*

- ***FRAME STRUCTURE***

***0xAA | ID | PAYLOAD | 0x55***

- *Synchronization delimiters*
- *Structured payload using a struct*
- *Optimized serial communication*

**UART SERIAL**

**TX** → **DATA**
**RX** ← **RESPONSE**

### *7.3 HYBRID COMMUNICATION*

The system employs **hybrid communication** via **Peer-to-Peer** for **efficient package transmission**, containing **positional data**, **robot states**, **game perception**, and **strategic information** for the dynamic development of plays. The architecture prioritizes **low latency**, continuous **synchronization** of collective positioning, **coordination** between agents, and **high reliability** in **decision-making** and **strategic** adaptation in real time.

*• ESP32 BOARD LOCATIONS*

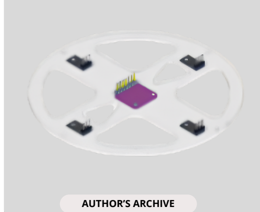
EYE → ESP 32 ON UPPER BOARD
HEAD → ESP 32 ON MAIN BOARD (ACTUATOR)
MUSCLE → ESP 32 ON MAIN BOARD (PROCESSOR)
FOOT → ESP 32 ON LOWER BOARD

## 8.0 SYSTEM TESTING AND VALIDATION

### *• COMPARISON OF TESTS AND IMPROVEMENTS BETWEEN ROBOTS*

#### *• 8.1 TESTS INVOLVING THE LOWER BOARD*

*FIGURE 18 - LINE SENSOR BOARD (2025 SEASON)*



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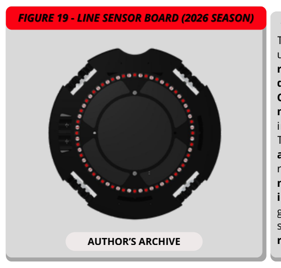
***• 2025 SEASON***

The **line board** from the **2025** season **(Figure 19)** was designed with **four QRE1113** sensors distributed **radially** at the four extremities of the robot, forming a **cross-shaped arrangement**. This configuration **expands detection coverage** through **optical sensing** based on **reflectivity**, allowing the line to be detected in **different directions**. The sensor arrangement provides **greater precision** in the robot's **positioning**, assisting with **alignment**, **trajectory corrections**, and **changes in direction**. The collected information enables **rapid responses** during **defensive movements** and **repositioning** during gameplay, contributing to **more precise** and **controlled** movements.

*FIGURE 19 - LINE SENSOR BOARD (2026 SEASON)*



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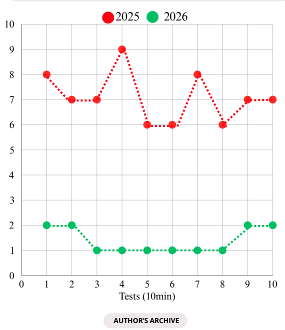
***• 2026 SEASON***

The **line board** from the **2026** season **(Figure 20)** uses **32 LEDs** and **32 LDRs**, providing **higher resolution** in line detection and allowing for a **more detailed reading** of the field surface. Two **CD74HC4067 multiplexers** perform **signal multiplexing**, enabling the sensors to be controlled in an **organized** and **efficient manner**.
This configuration improves **angular accuracy**, **alignment**, and the **speed of corrections** during the robot's movement, making its responses **faster** and **more stable**. In addition, the red LEDs enhance the **identification** of the white line in **contrast** with the green field, **facilitating the differentiation** between surfaces and contributing to a **more reliable reading**.

*FIGURE 20 – SCATTER PLOT COMPARING THE BOARDS*



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***• 9.1.2 SCATTER PLOT***

The graph presents the **comparison** between the **line sensor boards** used in the **2025** and **2026** seasons **(Figure 20)**, considering the **number of field exits** during the tests. **Ten** experimental sessions of 10 minutes were conducted for **each board**, totaling **100 minutes of testing** per model. The robots were tested alone on the field, focusing on compliance with the ***OUT BOUNDS rule***, aiming to **minimize exits** from the playing area.
The **2025 board** presented between **6** and **9** exits **per test**, while the **2026 board**, composed of 32 LEDs and 32 LDRs, maintained results between **1** and **2** exit occurrences. This reduction indicates **greater precision** in line detection and greater **stability** in trajectory corrections, contributing to **more consistent robot** control during movement.
***2025 BOARD:*** Higher number of exits;
***2026 BOARD:*** Lower number of exits.

#### *• 8.2 TESTS INVOLVING THE MIRROR*

*FIGURE 21 - CONVEX MIRROR*

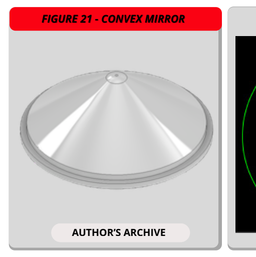

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*FIGURE 22 - VISION OF THE OPENMV INTERFACE*

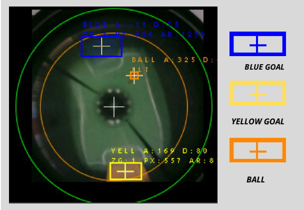

*FIGURE 23 - HYPERBOLIC MIRROR*

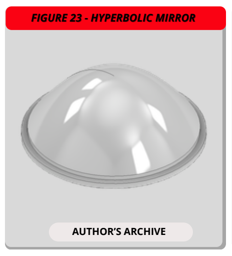

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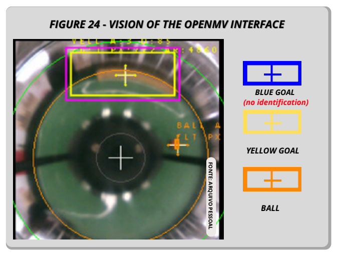
*FIGURE 24 - VISION OF THE OPENMV INTERFACE*



### *8.2.1 CONCLUSION*

**Two mirror geometries** were evaluated for the **omnidirectional vision system:** convex and hyperbolic. The convex mirror, **presented in Figure 21**, provides a wide reflection area, **allowing the OpenMV to obtain** a comprehensive view of the field, as **presented in Figure 22**. In this configuration, the system could simultaneously **identify 3 key field** elements: the **ball**, the **blue goal**, and the **yellow goal**, providing the information required for the robot's movement strategy and decision-making. The **hyperbolic mirror**, presented in **Figure 23**, has a more **specific geometry for image formation.** In the tested configuration, the view obtained by the **OpenMV**, presented in **Figure 24**, allowed the system to simultaneously identify only **2 key elements:** the **ball and the yellow goal.** The blue goal was not visible in the captured view, reducing the amount of strategic information available simultaneously to the vision system.
The comparison showed that the **convex mirror increased** the number of simultaneously identifiable strategic elements from **2 to 3,** representing **50%** more available visual information **than the hyperbolic** configuration. For the strategy developed by the team, this wider coverage allows the robot **to maintain a complete reference** of the ball and both goals in real time, facilitating the **selection of direction** and actions **during the match.** Therefore, the **convex mirror** was **selected as** the most suitable configuration, contributing to faster, continuous, and **more efficient decision-making.**

## 9.0 RESULTS

### *• COMPARISON OF TESTS AND ACCURACY BETWEEN ROBOTS*

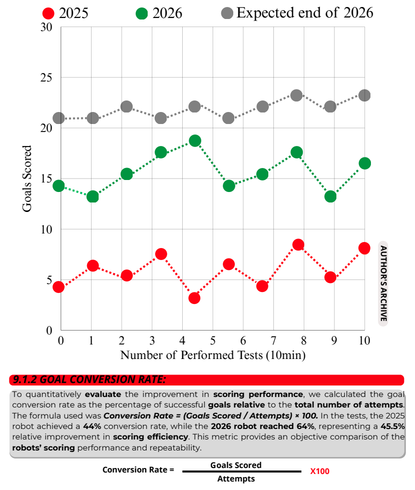
The **results obtained** demonstrate a **significant evolution in the system's offensive accuracy** between the **2025 and 2026 seasons.** The modifications **implemented in the electronic architecture**, **perception systems**, and **decision-making mechanisms** provided **measurable gains** in the precision of **offensive actions** and in **finishing efficiency**. The configuration **developed for 2026** demonstrated a greater capacity for **positioning**, better **alignment** with the ball, and **more consistent** responses during **dynamic game situations**.
In the **comparative analysis**, an **increase in the goal conversion rate**, greater **play repeatability**, and a **reduction in performance variations** throughout the matches were observed. The **2026** season robot demonstrated more **predictable** behavior, greater **operational stability**, and a **better ability to execute offensive actions** with **precision**.
Overall, the **results** demonstrate that the solutions adopted for 2026 not only **mitigated** limitations observed in 2025, but also **significantly increased** the level of **offensive efficiency** and **goal-scoring accuracy** of the system, consolidating a more **robust**, **precise**, and **competitive** architecture.

### *9.1 SCATTER PLOT*

- ***2025:*** Base system;
- ***2026:*** Optimized system;
- ***Expected end of 2026:*** Projected performance target (+20 goals/10 min).

*The graph (Figure 26) presents a quantitative comparison of the **number of goals scored during 10-minute tests** between the robots from the **2025 and 2026 seasons.** The tests were performed **without an opposing defender on the field**, allowing the scoring performance of each robot to be evaluated **under the same conditions.** The **red points represent the 2025** base kicking system, while the **green points correspond to the 2026 optimized robot**. In the tests, the 2025 robot achieved an average of **11 goals per 10-minute test**, while the 2026 robot **increased this average to 16 goals**, representing a **45%** improvement in scoring performance.*
*This improvement resulted from the optimization of the **kicking system and the development of a more adaptive attacking strategy,** allowing the robot to make better use of **available scoring opportunities.** The **gray expectation line** represents the team's **target of 20 goals per 10-minute** match, serving as a reference for **future development**. Although the 2026 robot **has not yet reached this target**, the comparison demonstrates a clear and **measurable improvement over the 2025 system and establishes a quantitative baseline** for further optimization.*

*FIGURE 25– SCATTER PLOT COMPARING GOAL-SCORING RATES*

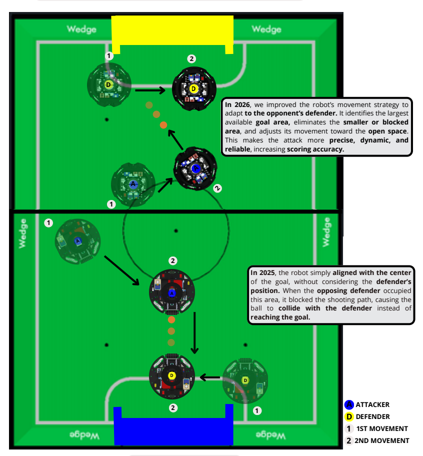


### *9.1.2 GOAL CONVERSION RATE:*

To quantitatively **evaluate** the improvement in **scoring performance**, we calculated the goal conversion rate as the percentage of successful **goals relative** to the **total number of attempts**. The formula used was ***Conversion Rate = (Goals Scored / Attempts) × 100.*** In the tests, the 2025 robot achieved a **44%** conversion rate, while the **2026 robot reached 64%**, representing a **45.5%** relative improvement in **scoring efficiency**. This metric provides an objective comparison of the **robots' scoring** performance and repeatability.

$$\text{Conversion Rate} = \frac{\text{Goals Scored}}{\text{Attempts}} \times 100$$

*FIGURE 26 – SIMULATIONS OF ROBOT MOVEMENTS*



In **2026**, we improved the robot's movement strategy to adapt **to the opponent's defender.** It identifies the largest available **goal area**, eliminates the **smaller or blocked area**, and adjusts its movement toward the **open space**. This makes the attack more **precise**, **dynamic**, and **reliable**, increasing **scoring accuracy.**

In **2025**, the robot simply **aligned with the center** of the goal, without considering the **defender's position.** When the **opposing defender** occupied this area, it blocked the shooting path, causing the ball to **collide with the defender** instead of **reaching the goal.**

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① ***Shooting trajectory of the 2026 season robot (Nexus).*** ② ***Shooting trajectory of the 2025 season robot (Fabuloso).***

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